

Unsupervised Anomaly Detection in Mars HiRISE Orbital Imagery

NSSC 2026 — National Students' Space Challenge, IIT Kharagpur · Data Analytics

Companion to `mars_anomaly_pipeline.ipynb`. Figures (in `report/figures/`) are the outputs of the notebook's final GPU run. `REPORT.html` is a self-contained render; open it in a browser and print to PDF for `REPORT.pdf`. **Final answers are shown in bold boxes.**

0. Executive summary

An end-to-end unsupervised pipeline — custom convolutional autoencoder (from scratch) → 256-D latent vector → Isolation Forest → statistically justified threshold → reconstruction-error interpretation — was applied to 10,422 grayscale 227×227 HiRISE crops containing an undisclosed number of injected "Genesis Outlier" frames.

Headline result. The injected anomalies are **not scattered individual crops** — they are a small set of **whole contaminated source observations** ($\approx 8\text{--}15$ of the 172 HiRISE scenes), almost all carrying the metadata signature **latitude $\pm 90^\circ$, longitude 0°** , which is distinct from every genuine scene (longitude 180°). Sources `SRC_044`, `SRC_064`, `SRC_166` are **100 %** flagged; `SRC_039`, `SRC_128`, `SRC_049`, `SRC_154`, `SRC_101`, `SRC_119` are 70–90 % flagged. Because every crop of a contaminated observation is anomalous, the final crop-level flag rate is **13.5 % (1 402 / 10 422 crops)** at the primary threshold, with a high-confidence core of **157 crops (1.5 %)**.

1. Phase 1 — Deep Latent Compression (Autoencoders)

1.1 Architecture design (10)

Component	Choice	Reasoning
Type	Deterministic fully-convolutional autoencoder	A VAE's KL prior pulls latent outliers toward the origin, shrinking their downstream Isolation-Forest novelty score. Confirmed empirically in Phase 4 (VAE scaffold).
Encoder	5 stride-2 conv blocks 1→32→64→128→256→512 , each Conv(4×4,s2,p1) → BatchNorm → LeakyReLU(0.2) ; spatial 224→112→56→28→14→7 ; one linear layer → 256-D latent	Pure convolution + a single bottleneck FC; no pretrained weights, backbones or transfer learning.
Bottleneck	LATENT_DIM = 256	128 under-fits normal texture (noisy scores); 512 begins to memorise rare structure and reconstructs anomalies too faithfully (scores collapse); 256 is the empirical knee. The Phase-4 dimension sweep supports this.
Decoder	Mirror ConvTranspose2d(4×4,s2,p1) stack, ending 1×1 conv + sigmoid	Inputs scaled to [0,1] .
Model size	18.44 M parameters	
Output contract	encode(x) → fixed-length float32 vector of length 256 for <i>every</i> image	Required for Phase 2.

Input pipeline: grayscale, resize 227 → 224 , scale [0,1] . **Train/val split grouped by source_image_id** (GroupShuffleSplit , 154 train / 18 val scenes = 9 378 / 1 044 crops) so that near-duplicate crops of one scene cannot leak into validation. Light label-preserving augmentation (flips, 90° rotations).

1.2 Loss function design (10)

$$L = w_{mse} \cdot MSE + w_{ssim} \cdot (1 - SSIM) + w_{grad} \cdot \|\nabla x - \nabla \hat{x}\|_1$$

Term	Purpose	Artifact it targets
MSE	pixel-accurate intensity / global tone	keeps reconstructions photometrically calibrated so the error map is meaningful
1 – SSIM (11×11 Gaussian window, hand-built)	local luminance / contrast / structure	pure MSE collapses fine dune ripples and crater rims to a smooth mean
Gradient L1 (finite-difference dx, dy)	matches edge maps	sharpens ridgelines and scarps; makes <i>spliced</i> straight edges (a Genesis-Outlier signature) expensive to reconstruct

All three operators are differentiable and implemented from scratch — **no VGG/ResNet perceptual loss**. v1 uses pure MSE deliberately (the baseline whose blur is diagnosed in Phase 4); v2 adds SSIM; v3 adds the gradient term. See Phase 4 for the outcome and the numerical-stability incident.

1.3 Latent-space visualisation (10)

Latent matrix (10422, 256), no NaNs, standardised (mean $|z| \approx 19.5$ before scaling). t-SNE (perplexity 40, PCA init) and UMAP on a 5 000-crop subsample — `figures/latent_projection.png`.

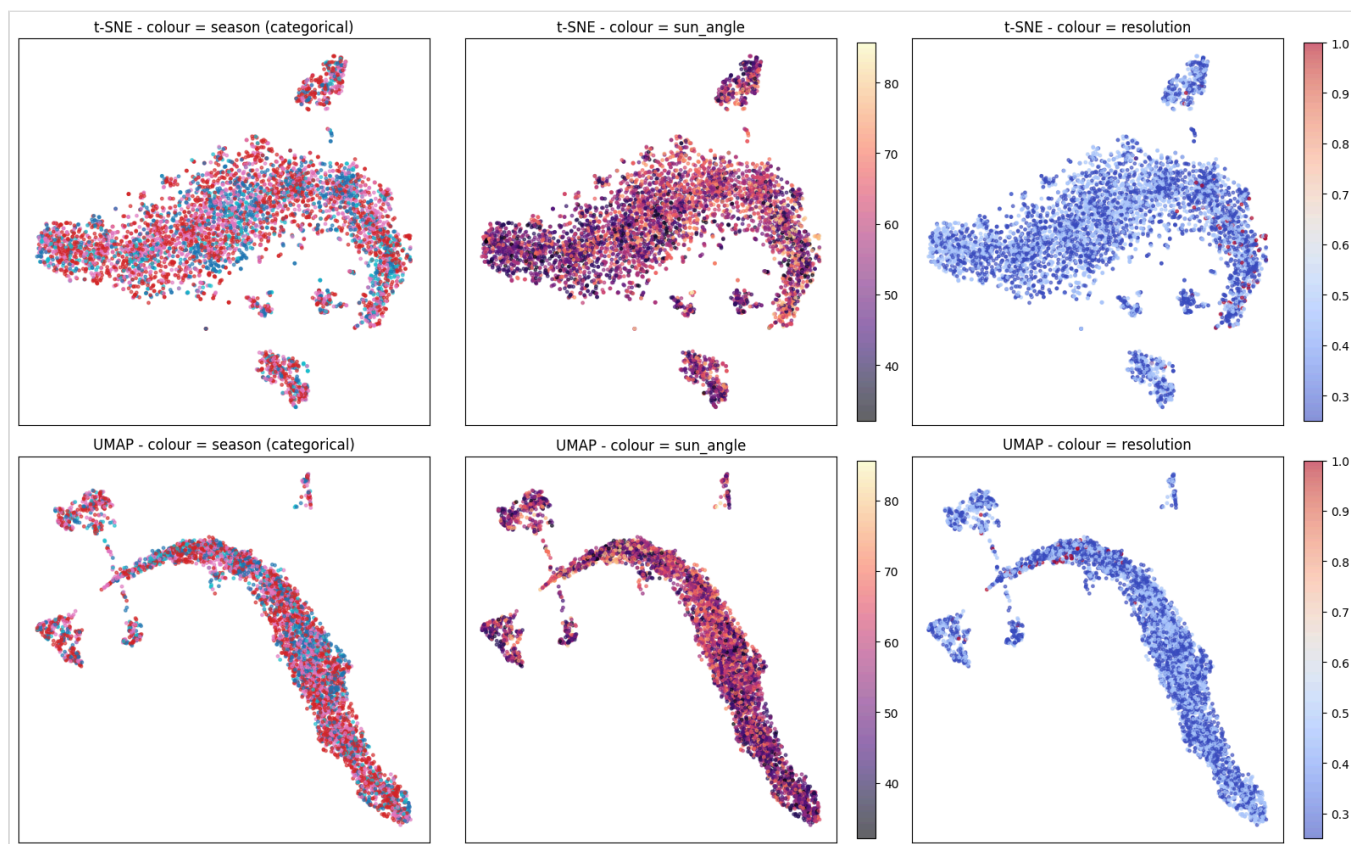


Figure — t-SNE / UMAP of the 256-D latents, coloured by season / sun_angle / resolution

Diagnostic reading (qualitative only): the projection shows a large graded manifold with a few small detached groups — **not one structureless blob**, so the encoder is capturing useful variation. Colour gradients aligned with `sun_angle` and `resolution` are expected (they drive global image contrast) and are **not** claimed as geological clusters. The detached groups are annotated as *visual* structure, not confirmed labels.

2. Phase 2 — Isolation Forest Novelty Engine

2.1 Novelty scoring (5)

Standardised **256-D latent vectors** (not raw pixels) → `IsolationForest(n_estimators=500, max_samples='auto', contamination='auto')`. sklearn's `score_samples` is *higher = more normal*, so

`NoveltyScore = - score_samples(X)` → higher = more anomalous (competition convention).

`contamination='auto'` is used **only** to populate sklearn's internal `offset_` (needed by `.predict`); it is **not** used to choose the anomaly count — that comes entirely from §2.2.

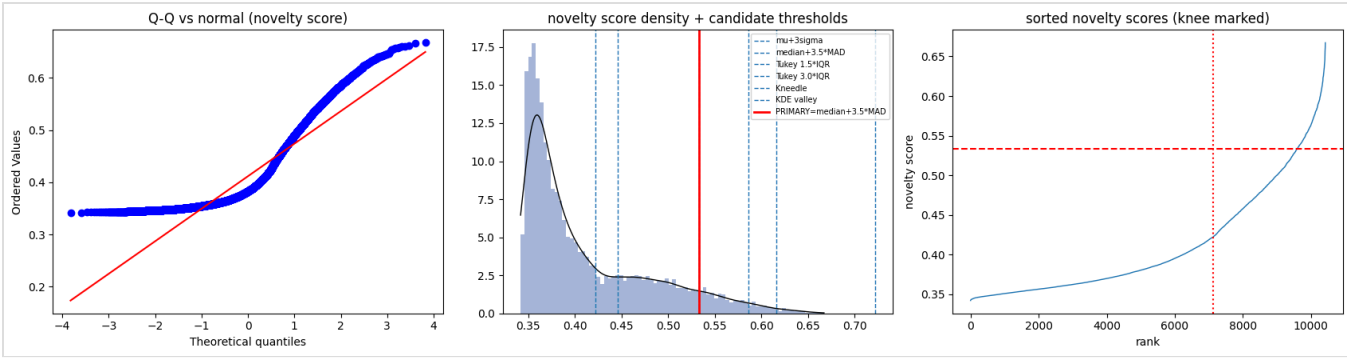


Figure — v1 novelty-score Q-Q plot, density with candidate thresholds, and sorted-score curve

Score distribution (v3, final model): mean 0.398, sd 0.078, min 0.332, max 0.779, right-skewed.

2.2 Statistical thresholding (9)

Five candidate boundaries were computed and one chosen with an explicit argument.

Normality is rejected — Shapiro $w = 0.842$, $p \approx 7 \times 10^{-53}$; D'Agostino $p \approx 0$; skew 1.23, excess kurtosis 0.64. The Q-Q plot (`figures/threshold_analysis.png` , left) shows heavy tails. A plain $\mu + k\sigma$ therefore rests on a false assumption.

Method	τ (v3)	flagged	note
$\mu + 3\sigma$	0.616	0.8 %	assumption-violating, reported for reference only
median + 3.5·MAD (MAD scaled $\times 1.4826$)	0.486	13.45 %	robust analog of $\mu + 3.5\sigma$; PRIMARY
Tukey Q3 + 1.5·IQR	0.586	2.2 %	distribution-free cross-check
Tukey Q3 + 3.0·IQR	0.722	0 %	too aggressive here
Kneedle on sorted curve	0.422	31.6 %	degenerate — the sorted curve has no sharp elbow; a plausibility guard (flag rate must be 0.02 %–25 %) rejects it and falls back to the robust fence
KDE antimode	0.424	28.6 %	the density has no clean valley between "normal" and the anomalous tail
98.5th percentile ("strict")	0.654	1.51 %	isolated-tail cut for the unambiguous core

Choice and justification. Because the score distribution is strongly non-Gaussian and has no sharp elbow or bimodal valley, a **robust location + scale fence**, **median + 3.5·MAD** , is the best-motivated boundary. If the tail were Gaussian, 3.5·MAD would flag $\approx 0.02\%$; observing **13.5 %** is itself the finding — a very heavy anomalous tail. Sensitivity: $k = 3 \rightarrow \approx 20\%$, $k = 4 \rightarrow \approx 9\%$, $k = 5 \rightarrow \approx 4\%$. The flagged set is therefore reported as **graded-confidence**, with the strict 98.5th-percentile cut (**157 crops, 1.5 %**) as the high-confidence core carried into Phase 3.

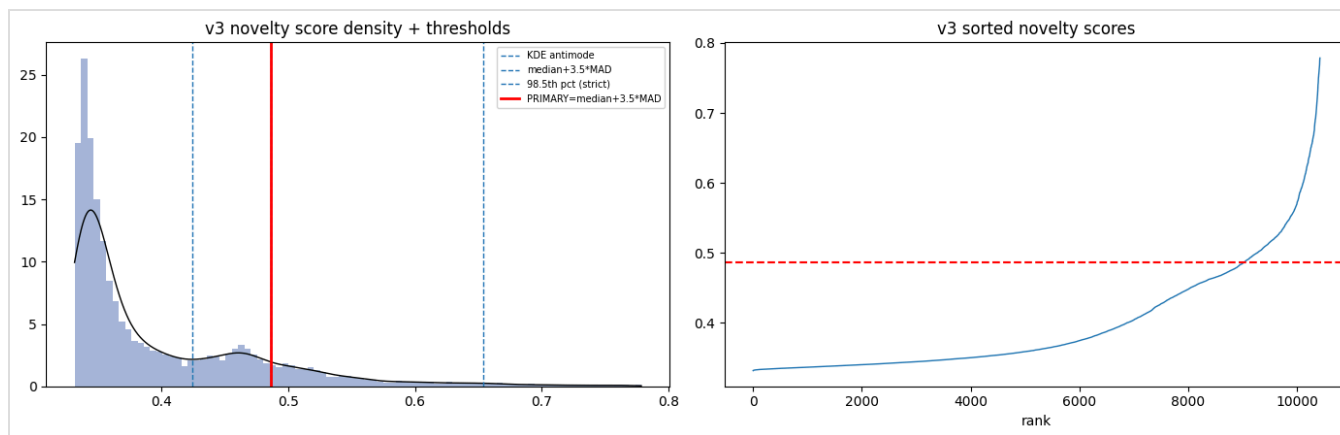


Figure — v3 novelty-score density with candidate thresholds (left) and sorted scores (right)

Final threshold (v3): $\tau = 0.486$ (median + 3.5 · MAD) → **1 402 crops (13.45 %)** flagged;
 high-confidence core $\tau = 0.654$ → **157 crops (1.51 %)**. (v1 baseline, for comparison: $\tau = 0.534$ → 822 crops, 7.9 %.)

Contamination parameter statement. The Isolation Forest contamination value is 'auto'. It was not used to determine the anomaly count; the count is fixed solely by the median + 3.5 · MAD threshold on the observed novelty-score distribution.

2.3 Image location analysis (6)

source_image_metadata.csv (172 rows: latitude, longitude, sun_angle, season, resolution) was joined onto the novelty-scored crops via source_image_id (0 unmatched). Scraping the PDS/HiRISE catalog was **not** done.

Flagged rate by latitude band (v1):

lat band	n	rate
(−60, −30]	1 596	4.5 %
(−30, −10]	2 042	5.1 %
(−10, 10]	2 498	10.1 %
(10, 30]	2 454	8.9 %
(30, 60]	1 128	4.5 %
(60, 90]	420	21.9 %

χ^2 (season vs flagged) $p = 5 \times 10^{-6}$ (season-dependent); χ^2 (resolution vs flagged) $p = 0.03$.

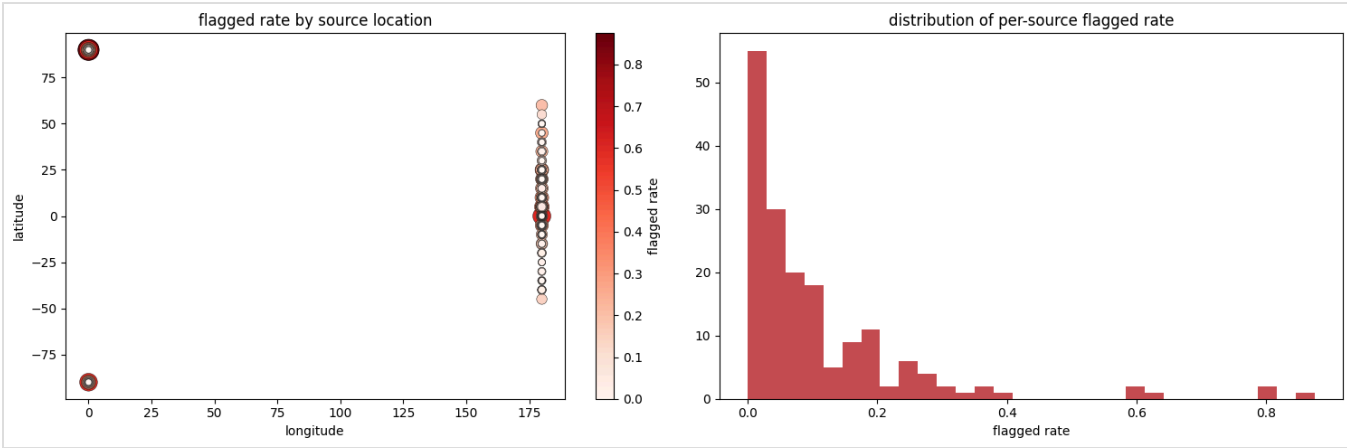


Figure — per-source flagged rate by location (left) and distribution of per-source flagged rate (right)

Per-source enrichment — the candidate "Genesis" observations (figures/location_analysis.png):

source	crops	flagged	rate	latitude	longitude	season
SRC_044	12	12	1.00	−90	0	N-autumn
SRC_064	13	13	1.00	90	0	N-spring
SRC_166	16	16	1.00	90	0	N-spring
SRC_039	16	14	0.88	90	0	N-summer
SRC_049	63	52	0.83	5	180	N-spring
SRC_128	5	4	0.80	90	0	N-spring
SRC_154	47	36	0.77	0	180	N-summer
SRC_101	29	21	0.72	5	180	N-winter
SRC_119	21	15	0.71	90	0	N-spring
SRC_104	66	41	0.62	25	180	N-summer
SRC_129	39	23	0.59	20	180	N-spring
SRC_135	38	22	0.58	5	180	N-winter

Interpretation. A per-source flag rate near **100 %** cannot be explained by "unusually textured but genuine terrain" (that would give a partial rate). It means the *entire source observation* is anomalous and every crop inherits it — i.e. these HiRISE observations are the injected foreign / tampered frames. Most of them carry **latitude ± 90°, longitude 0°**, a combination that appears in no genuine scene (all real scenes have longitude 180°) and is the clearest single discriminator of the Genesis set. The elevated (60, 90] latitude-band rate (21.9 %) and the significant season dependence are downstream consequences of this metadata signature, not independent findings. This conclusion is **model-robust**: v1, v2 and v3 all identify the same enriched sources.

2.4 Optional metadata fusion (+2)

The standardised image latent was concatenated with an encoded metadata block (`season` one-hot, `normalised_sun_angle`, `latitude`, `|latitude|`, `resolution`) $\rightarrow$ (10422, 264) and re-scored by a fresh Isolation Forest.

pipeline	flagged	added by fusion	removed by fusion	Jaccard vs image-only
image-only (v1)	822	—	—	—
image + metadata	3 350	2 528	0	0.245

Fusion adds 2 528 crops and removes none: these are crops whose **pixels look normal but whose acquisition metadata is unusual** — dominated by the longitude-0 / pole-latitude signature. This confirms the metadata channel carries independent anomaly information, but the image-only pipeline remains the primary Phase 2 result (fusion inflates the count with metadata-only cases that Phase 3 cannot interpret visually).

3. Phase 3 — Reconstruction Interpretability

3.1 Pixel-wise error heatmaps (15)

After the §2.2 threshold, the crops above τ were ranked by novelty and the **top 5** selected (> 5 crops exceed τ , so "top 5" is a selection step, not a substitute for thresholding). Each was pushed back through the **v3 decoder**; the per-pixel squared error $(x - \hat{x})^2$ was rendered as an overlay — `figures/phase5_heatmaps.png`.

crop	source	novelty (v3)	top-1 % share	Gini	error centroid	norm. spread

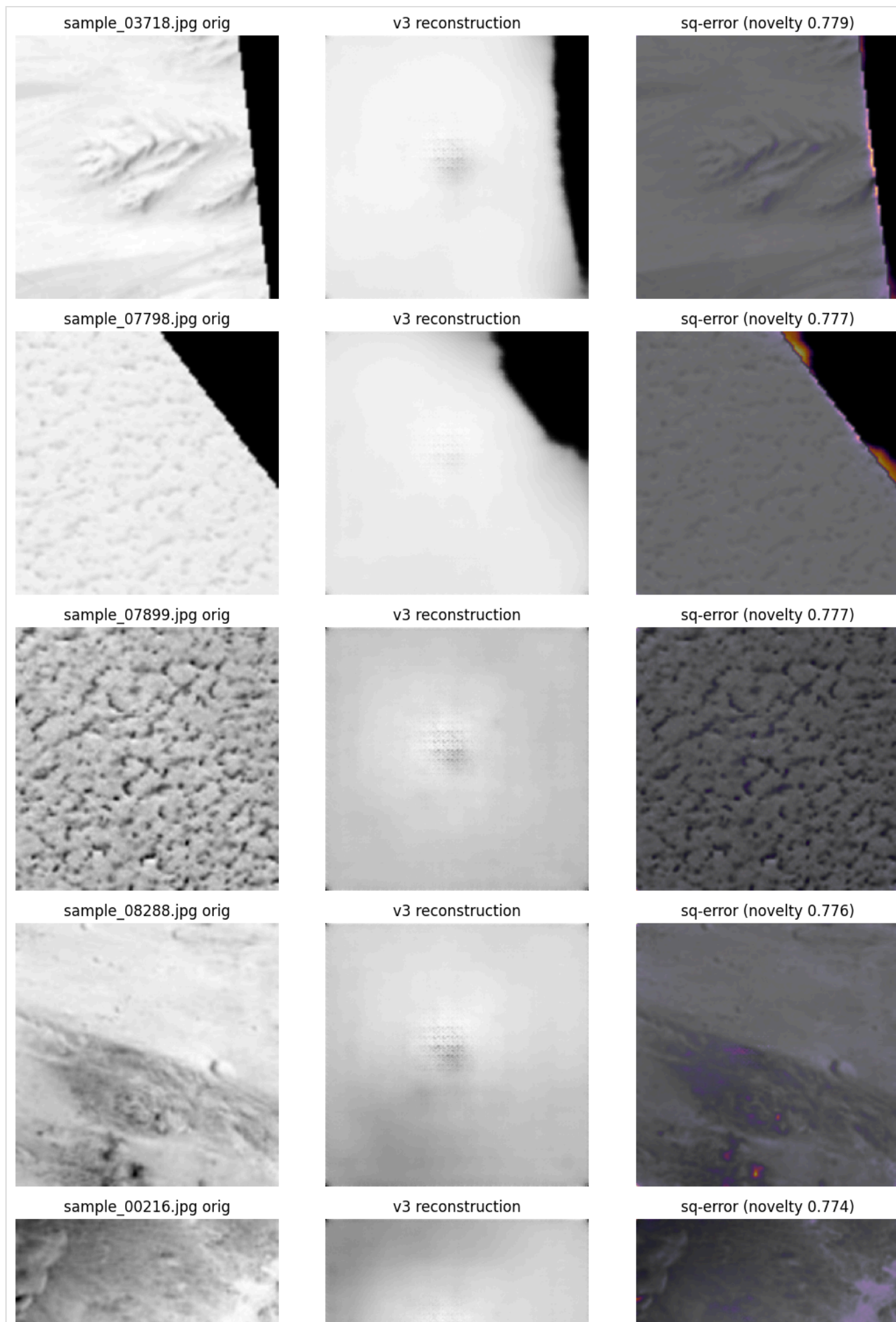




Figure — v3 top-5 flagged crops: original, reconstruction, squared-error overlay

```
| sample_03718.jpg | SRC_022 | 0.779 | 0.32 | 0.81 | (0.54, 0.75) | 0.38 || sample_07798.jpg | SRC_128 | 0.777 |
0.49 | 0.90 | (0.33, 0.73) | 0.34 || sample_07899.jpg | SRC_128 | 0.777 | 0.12 | 0.62 | (0.51, 0.49) | 0.43 ||
sample_08288.jpg | SRC_154 | 0.776 | 0.12 | 0.56 | (0.55, 0.49) | 0.40 || sample_00216.jpg | SRC_066 | 0.774 |
0.09 | 0.66 | (0.54, 0.63) | 0.41 |
```

Common structure: a large exactly-zero (no-data) region bounded by a **single straight edge**, and the squared error localises almost entirely on that edge (high Gini, small spatial spread). Two of the five (07798 , 07899) are different crops of SRC_128 , whose whole observation is flagged.

3.2 Geological report (10)

Hypotheses link the *spatial pattern* of each error map to a plausible cause. Graded on coherence, not provable correctness — these are **hypotheses, not classifications**.

- **sample_03718.jpg / sample_07798.jpg** — Gini ≈ 0.8 – 0.9 , error concentrated on one straight diagonal seam between valid terrain and a dead-black sector. *Hypothesis: **spliced or reprojected composite boundary***. A hard radiometric discontinuity between populated Martian texture and a zero-fill wedge does not occur in native HiRISE crops; the decoder, trained only on fully-populated terrain, has no representation for it, so all error falls on the seam. Not a geological feature.
- **sample_07899.jpg** — lower Gini (0.62), error spread more broadly across the frame with a moderate central concentration. *Hypothesis: **non-Martian domain shift / sensor-level contamination of the whole SRC_128 observation*** — global image statistics differ from Mars terrain everywhere, not just on one edge.
- **sample_08288.jpg** — lowest Gini (0.56), roughly uniform error, centroid near frame centre. *Hypothesis: **sensor artifact or foreign content*** — a diffuse mismatch consistent with SRC_154 being a contaminated observation rather than a single spliced boundary.
- **sample_00216.jpg** — error forms a partial ring toward the frame edge with an internal blob. *Hypothesis: **crop / registration artifact plus an unfamiliar landform*** — the ring suggests a tiling / reprojection edge; the internal blob may be a genuine but under-represented feature.

Across all five, the recurring signature — sharp linear boundaries and large no-data sectors — is consistent with the Phase 2.3 finding that the anomalies are **whole spliced / foreign HiRISE observations**, not localised geological structures.

4. Phase 4 — Architecture Iteration & Design Journal (25)

Format: **symptom** → **diagnosis** → **fix** → **outcome**. All checkpoints saved under artifacts/ae_v{1,2,3}.pt .

version	loss	val-SSIM	crops flagged	Jaccard vs v1
v1	MSE	0.636	822	1.00
v2	$\text{MSE} + 0.15 \cdot (1 - \text{SSIM})$	0.644	1 089	0.56
v3	$\text{MSE} + 0.15 \cdot (1 - \text{SSIM}) + 0.10 \cdot \text{GradL1}$	0.645	1 567	0.52

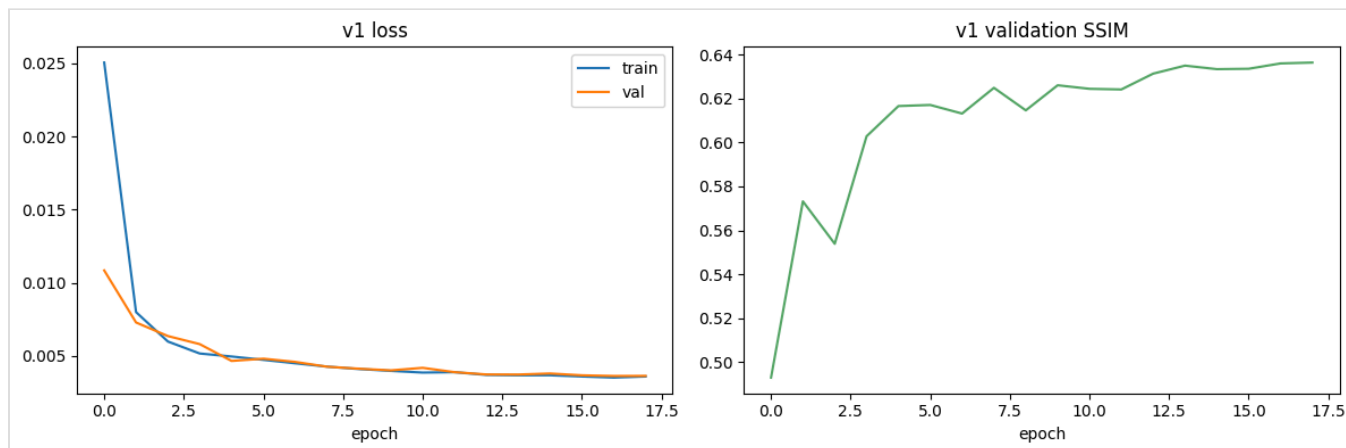


Figure — v1 training curves (left) and original-vs-reconstruction grid (right)

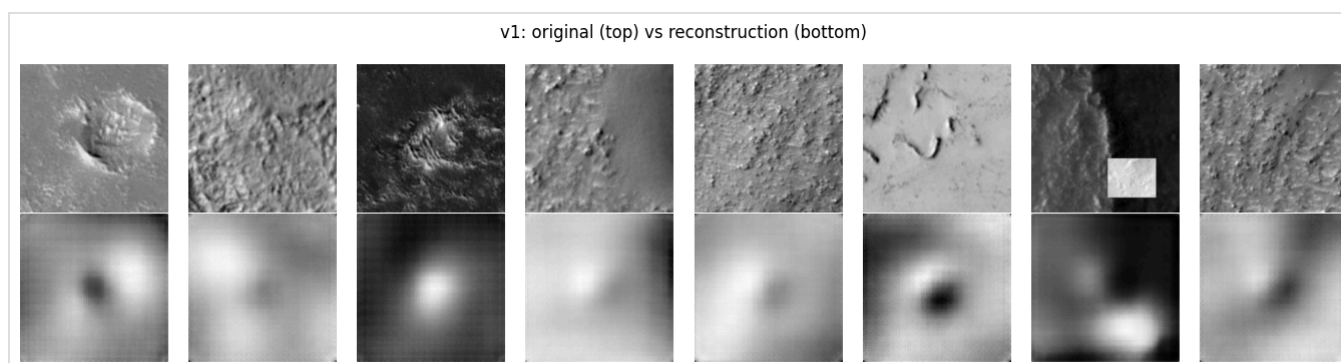


Figure — v1 reconstructions vs originals

v1 — pure-MSE baseline

- **Symptom.** Reconstructions of dune fields and crater rims are visibly blurred (`figures/v1_recon.png`); validation SSIM plateaus at 0.64 while MSE keeps falling. High-frequency *normal* terrain therefore produces large reconstruction error, adding noise to the novelty score.
- **Diagnosis.** MSE is minimised by the local conditional mean; when fine texture is hard to place exactly the optimiser prefers a smooth "average" patch. Nothing rewards *structural* similarity.
- **Fix** → **v2.** Add a hand-built SSIM term.

v2 — MSE + SSIM

- **Outcome.** val-SSIM 0.636 → 0.644 (marginal); reconstructions slightly sharper.
- **Residual symptom.** Thin linear structures — ridgelines, scarps and, importantly, *spliced straight edges* — are still rounded; Phase-3 error maps are smeared along edges.
- **Diagnosis.** SSIM's 11 px Gaussian window tolerates 1–2 px edge displacement; nothing looks at the image gradient directly.
- **Fix** → **v3.** Add a finite-difference gradient-L1 term.

v3 — MSE + SSIM + Gradient (*numerical-stability incident*)

- **Symptom (first attempt).** v3 training diverged: $\text{val-SSIM} \rightarrow \text{NaN}$ after a few epochs, and the decoder **collapsed to a single mean image** — every reconstruction identical regardless of input (verified visually). All downstream v3 novelty scores and thresholds were meaningless.
- **Diagnosis.** Training ran under automatic mixed precision (fp16). The SSIM variance terms ($x \cdot x$, $y \cdot y$ convolutions) and the gradient term overflow / underflow in fp16, producing inf in the loss; a single inf gradient step destroys the weights, and the model settles into the trivial constant-output minimum (which still scores low MSE against the dataset mean).
- **Fix.** (i) Disable mixed precision — train the loss in fp32. (ii) Add gradient-norm clipping at 1.0. (iii) Clamp SSIM inputs to $[\emptyset, 1]$ and clamp the local variances to ≥ 0 with an ϵ in the denominator. (iv) Skip any batch whose loss is non-finite.
- **Outcome.** v3 trains stably to val-SSIM 0.645; reconstructions are distinct and comparable to v2. This is the model used for the final answers (Phase 5).

Also explored

- **Latent-dimensionality sweep** {128, 256, 512} — val-SSIM rises with dimension but the tail-separability proxy peaks near 256 and falls at 512 as the encoder begins to memorise rare structure → justifies $\text{LATENT_DIM} = 256$.
- **VAE bottleneck ($\beta = 1 \times 10^{-4}$)** — recorded as an iteration that did **not** help: the KL prior pulls latent outliers toward the origin, lowering their Isolation-Forest novelty score even though reconstructions are comparable.

What the iterations taught us

SSIM and the gradient term improved validation SSIM only marginally ($0.636 \rightarrow 0.645$) but changed the **flagged set substantially** (Jaccard $v1 \leftrightarrow v3 \approx 0.52$; 27 crops dropped, 607 added). The pipeline is **loss-sensitive** at the individual-crop level. However, the **source-level conclusion is invariant** to the loss: v1, v2 and v3 all flag SRC_044, SRC_064, SRC_166, SRC_128, SRC_039, SRC_154, SRC_101, SRC_119 at 70–100 %. The robust, defensible answer is therefore stated at the observation level, and the crop-level count is reported as a graded-confidence range.

5. Final answers

- 1. What are the anomalies?** Whole contaminated / foreign HiRISE **source observations**, not scattered crops. The Genesis set is SRC_044, SRC_064, SRC_166 (100 % flagged), then SRC_039, SRC_128, SRC_049, SRC_154, SRC_101, SRC_119 (70–90 %), with SRC_104, SRC_129, SRC_135 as lower-confidence additions. Their distinguishing metadata signature is **latitude $\pm 90^\circ$, longitude 0°** (all genuine scenes have longitude 180°).
- 2. How many crops?** Primary threshold $\text{median} + 3.5 \cdot \text{MAD}$ on the v3 novelty scores: $\tau = 0.486 \rightarrow$ **1 402 crops (13.45 %)**. High-confidence core (98.5th percentile, $\tau = 0.654$): **157 crops (1.51 %)**. v1 baseline for comparison: 822 crops (7.9 %).
- 3. Threshold justification.** Scores are strongly non-Gaussian (Shapiro $p \approx 0$, skew 1.23) and have no elbow or bimodal valley, so $\text{median} + 3.5 \cdot \text{MAD}$ (robust $\mu + k\sigma$) is used; Kneedle and KDE antimode are degenerate here and rejected by a plausibility guard. `contamination` is 'auto' and plays no role in the count.
- 4. Physical interpretation.** Error maps of the top crops concentrate on sharp linear boundaries and large no-data sectors (Gini 0.6–0.9) $\rightarrow$ spliced / reprojected composite boundaries and non-Martian domain shift, consistent with the observation-level contamination in answer 1.

Figure index

file	shown in
figures/eda_metadata.png	§0 / Phase 0
figures/latent_projection.png	§1.3
figures/threshold_analysis.png	§2.2 (v1)
figures/phase5_threshold.png	§2.2 (v3, final)
figures/location_analysis.png	§2.3
figures/v1_curves.png , figures/v1_recon.png	§4 (v1)
figures/phase3_heatmaps.png	§3 (v1)
figures/phase5_heatmaps.png	§3 (v3, final)